Midterm Project: Dataset Description

1. General Information

Dataset Title	Marketing Campaign Dataset
Dataset Source	Kaggle (rodsaldanha/arketing-campaign)
Business Domain	Marketing / Customer Analytics
Dimensions	2,240 rows and 29 columns

2. Business Problem & Analytical Objective

For this project, I am focusing on customer analytics. The main objective is to look at how different types of customers interact with a company's marketing campaigns and where they spend their money. By doing this, I want to find out which customer groups are the most profitable and how the business can shift from mass marketing to targeted marketing. Getting a better understanding of these buying patterns should ultimately help the company improve its marketing ROI.

3. Dataset Justification

I chose the "Marketing Campaign" dataset because it fits perfectly with the midterm requirements. With 2,240 rows and 29 columns, it's well over the 1,000 row and 8 column minimum. It also has a really good mix of categorical data (like education and marital status) and numerical data (like income and amount spent on products).

Most importantly, the data isn't perfectly clean. I already noticed it has some real-world messy parts, like missing incomes and some obvious outliers in the birth year column. This will give me a great opportunity to demonstrate my data cleaning skills in Excel before I bring the data into Power BI to build the dashboard.

4. Description of Variables

Here is a breakdown of the 29 columns in the dataset, grouped by category:

Customer Demographics

- **ID:** Unique number for each customer.
- **Year_Birth:** The year the customer was born.

- **Education:** Highest level of education completed.
- **Marital_Status:** Current relationship status.
- **Income:** Annual household income.
- **Kidhome:** Number of small children living at home.
- **Teenhome:** Number of teenagers living at home.

History & General Behavior

- **Dt_Customer:** The date they first became a customer.
- **Recency:** How many days it has been since they last bought something.
- **Complain:** Shows as a 1 if they complained in the last 2 years, otherwise 0.

Spending Habits (Last 2 Years)

- **MntWines:** Money spent on wine.
- **MntFruits:** Money spent on fruit.
- **MntMeatProducts:** Money spent on meat.
- **MntFishProducts:** Money spent on fish.
- **MntSweetProducts:** Money spent on sweets.
- **MntGoldProds:** Money spent on premium/gold products.

Channels & Campaign Responses

- **NumDealsPurchases:** Purchases made using a discount.
- **NumWebPurchases:** Purchases made on the website.
- **NumCatalogPurchases:** Purchases made from a catalog.
- **NumStorePurchases:** Purchases made in physical stores.
- **NumWebVisitsMonth:** Website visits in the last month.
- **AcceptedCmp[1-5]:** Five separate columns showing if the customer accepted offers in campaigns 1 through 5 (1 = yes, 0 = no).
- **Response:** Shows a 1 if they accepted the offer in the most recent campaign, 0 if not.

Note: There are two extra columns (Z_CostContact and Z_Revenue) that contain standard structural data from the source and won't be heavily used in the analysis.